

- 1 (a) To hit the coconut horizontally, vertical component of the velocity = 0.
Initial vertical velocity = $20 \sin \theta$.

Using $v^2 = u^2 + 2as$, and setting v to 0 m s^{-1} and $s = 18.0 - 2.2 = 15.8 \text{ m}$:

$$0 = (20 \sin \theta)^2 - 2(9.81)(15.8)$$

$$\sin \theta = 0.880$$

$$\theta = 61.7^\circ$$

- (b) By $v = u + at$,

$$0 = 20 (0.880) - 9.81 \times t$$

$$t = 1.79 \text{ s}$$

- (c) Horizontal displacement = $20 \times \cos(61.7^\circ) \times 1.79$
= 17.0 m

- (d) The downward acceleration will be larger and hence the same initial vertical velocity will be reduced to zero over a shorter vertical displacement.

Hence, θ has to be larger so that the initial vertical velocity is larger. Since initial horizontal velocity is smaller and it will decrease due to the air resistance in the horizontal direction, the horizontal displacement will be lower.

- 2 (a) product of mass and (linear) velocity